GREENSEO · THE WALRUS, BRIGHTON · 7 OCTOBER 2026

Influence for impact

Getting sustainable AI past the C-suite

Ed Chinn · Founder, Kintara



THE GAP · 02

Hands up



Who has been asked to do more with AI this year?



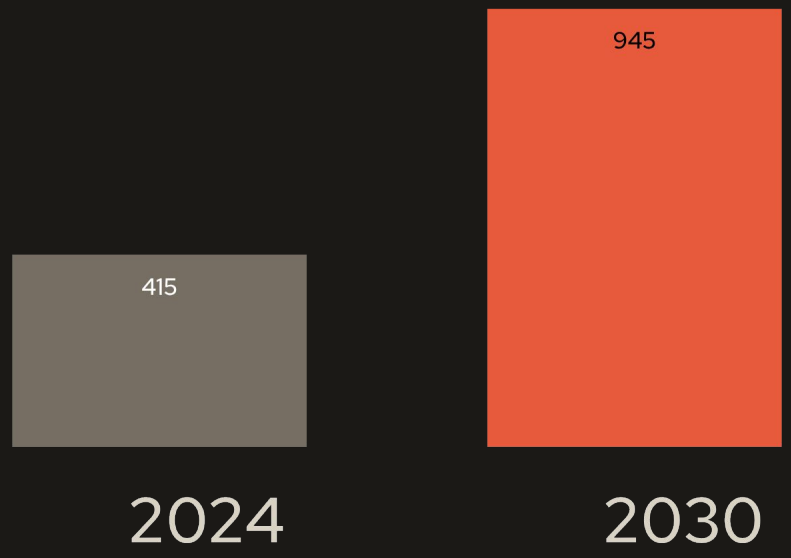
Who has been asked what it costs the planet?

Tonight: closing the gap between those two sets of hands.

THE GAP · 03

The big number

Data centre electricity, TWh



415 → 945 TWh a year

Data centre electricity, 2024 to 2030. More than Japan uses today.

Most of AI's water is used upstream, in the power stations making its electricity, not in cooling the servers.

Sources: IEA, Energy and AI (2025, updated 2026); researcher estimates via Nina Benoît, AI4G webinar, July 2026

THE GAP · 04

The small number

Grams of CO_2e

A text prompt (Google Gemini): **0.03 g**

A web page view (average page): **0.36 g**

A longer answer (Mistral Large 2): **1.14 g**



One text prompt ≈ 9 seconds of TV ≈ 5 drops of water

Same kind of request, ≈40× apart. It depends on the model and on what gets counted.

It's not the prompt. It's the pattern.

Sources: Google, median Gemini text prompt, 0.24 Wh, market-based (Aug 2025); Mistral AI life-cycle analysis (Jul 2025); Website Carbon Calculator

THE GAP · 05

Swap the argument that loses

LOSES:

~~"AI is bad for the planet. Please use less."~~

GETS FUNDED:

"Let's cut waste, risk and cost."

You've won this argument before:

31% better LCP → 8% more sales.

Lighter pages use less energy too. The bonus, not the pitch.

You already hold the keys: you measure what nobody looks at, find the waste and prove it in numbers a director accepts.

Source: web.dev, Vodafone case study: a 31% improvement in LCP increased sales by 8%

THE GAP · 06

Adoption has run ahead of results

88%

of organisations use AI in at least one function

$500bn+

corporate AI investment in 2025

74%

of companies have yet to show tangible value from AI

What wastes money in AI and what wastes energy in AI are the same thing.

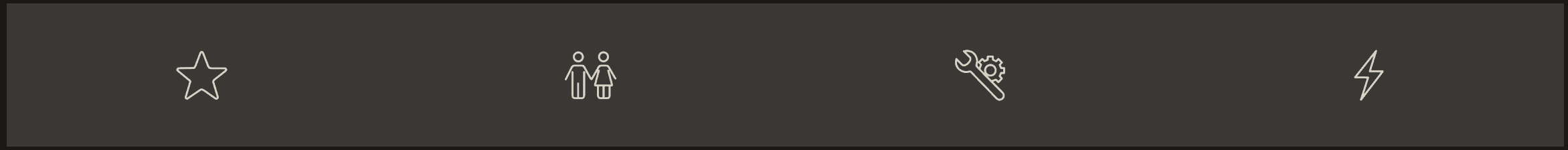
Sources: McKinsey, The state of AI (November 2025); Stanford AI Index 2026, via Kintara, From AI adoption to AI stewardship (June 2026); BCG, Where's the value in AI? (October 2024), 1,000 executives in 59 countries

THE FIX · 07

Flip the order

Usually: pick a shiny model → find a use for it → worry about governance later

Frugal AI:



Goal: what are we trying to achieve?

Workflow: where it fits, who checks it

Model: the smallest that does the job

Compute: where it runs, on what power

Frugal isn't cheap. It's more with less, on purpose.

Drawing on frugal AI thinking, including the Frugal AI Hub at Cambridge Judge Business School

THE FIX · 08

Drop, beaker, bucket



A drop: simple text tasks. Bulk meta descriptions, a regex, structured markup.



A beaker: complex reasoning. Search intent analysis, clustering competitor gaps.



A bucket: heavy media and agents. Video, images at scale, an agent crawling sites.

TURN TO YOUR NEIGHBOUR · 30 SECONDS

Which AI tool in your stack is a bucket doing a drop's job?

Drop, beaker, bucket: with thanks to Dr Malcolm Buckley, co-chair, CISL Alumni AI for Good

THE FIX · 09

Three levers you can pull



01 Right-size

A small, task-specific model by default. Ask when you need a bigger one.

Up to 90% less energy. Shorter prompts and answers: over 50% less.



02 Location

Ask where it runs and on what power. Ask for location-based figures, not just market-based.

Google's 0.03 g per prompt is market-based.



03 Duplication

Three tools built in an afternoon. Or one, built once, with guardrails.

Sources: UNESCO and UCL, Smarter, smaller, stronger (Jul 2025); Luccioni et al., Power Hungry Processing (FAccT 2024); Google, Gemini inference technical paper (Aug 2025)

THE FIX · 10

Give them three options

Option 1 · Carry on: licences, tools and footprint as they are.

OPTION 2 · RECOMMEND THIS ONE

List, consolidate, report: what we pay for and use. Fewer, right-sized tools with guardrails. Cost, usage and footprint in a quarter.

Option 3 · Audit and rebuild: every AI workflow, from scratch.



The compromise effect: offered three, people pick the middle option more often than they would on its own.



Bring one surprise: three tools doing one job, or licences nobody has opened since March.

An illustrative composite of a fifty-person agency. Compromise effect: Simonson, Journal of Consumer Research (1989)

The Adoption Iceberg

Above the waterline · ≈20% of value

Licence, platform, launch. What most budgets fund.

Below the waterline · ≈80% of value

Workflow redesign · skills · change and culture · governance and feedback · deciding what AI is not for

The ask isn't spend less. It's spend differently.

Sources: Kintara, From AI adoption to AI stewardship (June 2026); BCG, Where's the value in AI? (October 2024): about 70% of AI challenges are people and process, 10% the algorithm

Who are you persuading?

WHICH ONE IS YOUR BOSS?

Don't ask how to convince them you're right. Ask what decision they need to make.

Head of marketing ·	instead of "carbon footprint" ·	say faster pages, better experience, more visibility
Finance director ·	instead of "sustainability" ·	say cost per outcome, licences we don't use
CTO and developers ·	instead of "green code" ·	say compute efficiency, lower latency
Your client ·	instead of "net zero" ·	say risk and reputation

The one swap to remember: "use less AI" → "get more from the AI we already pay for"

When they ask about AI's footprint

Owning up costs a little trust. Being found out costs a lot more.

A label gives them nothing to check: "We're committed to responsible AI."

An explanation gives them something to check: "It runs in [region] on [grid]. A named person checks every output. Here are the numbers, and our sources."

01

Acknowledge: yes, it has a footprint. Here's what we know, and what we don't.

02

Explain: where it runs, what it replaces, who's accountable.

03

Show: the numbers in context, with sources and the gaps named.

Research: Schilke and Reimann, The transparency dilemma, Organizational Behavior and Human Decision Processes 188 (2025)

Your one-page case

1. The problem, in one sentence
2. What it costs now: money, time, footprint
3. The frugal option: right-sized, guarded
4. What we'll measure, and when
5. What we won't use AI for



Readable while the kettle boils.

Line 5 surprises leaders, and builds a lot of trust.

Reflective motivation: the COM-B model of behaviour change (Michie et al., 2011)

MONDAY · 15

On Monday, pick one

1

Start a baseline: none of the 17 organisations in our research had one. A rough list of tools puts you ahead.

2

Ask one supplier: where does it run, on what power? Can we have location-based figures?

3

Tag it, celebrate it: a new habit took a median of 66 days to stick. Tag the work, count it, celebrate it.

Tell the person next to you which one you'll do.

Baseline: Kintara, From AI adoption to AI stewardship (June 2026). Habits: Lally et al., European Journal of Social Psychology (2010)

One action a month

Doing better with AI, not just more.

GreenSEO asks every member for one action a month. Take one seed home tonight:

ONE BASELINE · ONE QUESTION · ONE TAG

Ed Chinn · ed@kintara.uk · Questions now, then over to Jimmy



Resources to take home

AI Energy Score Leaderboard: compare the energy use of AI models on the same tasks

EcoLogits: estimate the footprint of calls to hosted AI models

Website Carbon Calculator and Ecograder: estimate the footprint of a web page

CO2.js: the open-source library behind many of these tools

Green Algorithms calculator: estimate the footprint of a computation

Smarter, smaller, stronger: UNESCO and UCL (2025), the research behind right-sizing

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The one-page case template is at the back of these slides

APPENDIX

The one-page case: template

1 The problem, in one sentence

For example: "We pay for four AI tools that do one job."

2 What it costs now

Money: licences and usage per month. Time: hours lost to rework or duplicate tools. Footprint: a rough estimate, with the method named.

3 The frugal option

Which tools we keep. Which model size is the default. Who checks the output.

4 What we'll measure, and when

Cost per outcome, usage and footprint. We report back on [date].

5 What we won't use AI for

Name two or three things, and why.

Fits on one page. Readable while the kettle boils.

Thank you

Book a meeting

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